Week 8

Module 4 - B | Latent Variable Models

Expected Learning Outcomes

---------- Week 7 ----------

1: Concept of Clustering

2: Hard Clustering – K-means

3: Concept of Latent Variables

4: Expectation-Maximization (EM)

5: Soft Clustering – Mixture Models

---------- Week 8 ----------

6: Applications – EM for Gaussian Mixture Models (GMM)

7: Applications – EM for Document Clustering

Part 6: EM for Gaussian Mixture Models (GMM)

Recall Mixture Models

hard

easy

easy

$$\sum_{n=1}^{N} \boxed{\log p\left(\mathbf{x}_n \mid \theta\right)} = \sum_{n=1}^{N} \log \sum_{z_n} p\left(\mathbf{x}_n, z_n \mid \theta\right) = \sum_{n=1}^{N} \log \sum_{k=1}^{K} \boxed{p\left(\mathbf{x}_n \mid z_n = k, \theta\right)} \boxed{p\left(z_n = k \mid \theta\right)}$$

marginal likelihood of observed (incomplete) data

conditional likelihood of observed (incomplete) data; probability density function/mass function of the k-th cluster

prior of the k-th cluster

The model decomposes the overall density $p(\mathbf{x} \mid \theta)$ into a weighted linear combination of K components (densities) $p(\mathbf{x} \mid z = k, \theta)$.

$p(z = k \mid \theta)$ represents the PRIOR of the the k-th cluster (the probability that a randomly chosen data point was generated by the k-th component).

Recall Mixture Models

$$\sum_{n=1}^{N} \log p\left(\mathbf{x}_n \mid \theta\right) = \sum_{n=1}^{N} \log \sum_{z_n} p\left(\mathbf{x}_n, z_n \mid \theta\right) = \sum_{n=1}^{N} \log \sum_{k=1}^{K} p\left(\mathbf{x}_n \mid z_n = k, \theta\right) p\left(z_n = k \mid \theta\right)$$

Challenges for solving MLE with hidden variables:

1: The log of a sum

2: The labels of the data points are unknown, that is, the specific class or component from which each data point belongs is unknown. This makes the problem analytically intractable.

The hidden labels make the current problem a typical problem with incomplete data for which the EM algorithm is designed for !!!

Recall EM

Motivation (Refer to Pages 21 & 22 of Week 7):

For maximizing the complete data log-likelihood, you directly solve several MLE separately. But for maximizing the incomplete data log-likelihood, the log of sum formulation makes it difficult to optimize, and this is why EM is introduced.

High-level Concept (Refer to Page 25 of Week 7):

1: Iterative bound optimization: in each iteration, we construct the optimal surrogate function Q (acting as a tight lower-bound) in the E-step and maximize the Q function (M-step)

2: Estimating hidden variables to obtain the expected complete data in the E-step and solving standard MLE in the M-step.

3: Coordinate ascent

The E-step for the Mixture Models

Compute the responsibility of cluster k for generating data point n, estimated using the current parameter estimates $\boldsymbol{\theta}$.

For each n and k, where $n = 1, \ldots, N$ and $k = 1, \ldots, K$, we calculate the following posterior probability for $z_n = k$:

$$
\begin{aligned}
p\left(\mathbf{z}_n = k \mid \mathbf{x}_n, \boldsymbol{\theta}\right) &= \frac{p\left(\mathbf{x}_n \mid \mathbf{z}_n = k, \boldsymbol{\theta}\right) p\left(\mathbf{z}_n = k \mid \boldsymbol{\theta}\right)}{p\left(\mathbf{x}_n \mid \boldsymbol{\theta}\right)} \\
&= \frac{p\left(\mathbf{x}_n \mid \mathbf{z}_n = k, \boldsymbol{\theta}\right) p\left(\mathbf{z}_n = k \mid \boldsymbol{\theta}\right)}{\sum_{j=1}^{K} p\left(\mathbf{x}_j \mid \mathbf{z}_j = k, \boldsymbol{\theta}\right) p\left(\mathbf{z}_j = k \mid \boldsymbol{\theta}\right)} \\
&= \frac{p\left(\mathbf{x}_n \mid \mathbf{z}_n = k, \boldsymbol{\theta}\right) p\left(\mathbf{z}_n \mid \boldsymbol{\theta}_k\right)}{\sum_{k'} p\left(\mathbf{x}_n \mid \mathbf{z}_n = k', \boldsymbol{\theta}\right) p\left(\mathbf{z}_n \mid \boldsymbol{\theta}_{k'}\right)}
\end{aligned}
$$

The M-step for the Mixture Models

$$\sum_{n=1}^{N} \mathbb{E}_{\mathbf{z}_n}\left[\log p\left(\mathbf{x}_n, \mathbf{z}_n \mid \boldsymbol{\theta}\right)\right] = \sum_{n=1}^{N}\sum_{k=1}^{K} p\left(\mathbf{z}_n = k \mid \mathbf{x}_n, \boldsymbol{\theta}\right) \log\left[p\left(\mathbf{x}_n \mid \mathbf{z}_n = k, \boldsymbol{\theta}\right) p\left(\mathbf{z}_n = k \mid \boldsymbol{\theta}\right)\right]$$

$$= \sum_{n=1}^{N}\sum_{k=1}^{K} p\left(\mathbf{z}_n = k \mid \mathbf{x}_n, \boldsymbol{\theta}\right) \log p\left(\mathbf{x}_n \mid \mathbf{z}_n = k, \boldsymbol{\theta}\right)$$

$$+ \sum_{n=1}^{N}\sum_{k=1}^{K} p\left(\mathbf{z}_n = k \mid \mathbf{x}_n, \boldsymbol{\theta}\right) \log p\left(\mathbf{z}_n = k \mid \boldsymbol{\theta}\right)$$

Let $r_{nk} = p\left(\mathbf{z}_n = k \mid \mathbf{x}_n, \theta\right)$ and $\varphi_k = p\left(\mathbf{z}_n = k \mid \boldsymbol{\theta}\right)$

, then we can simplify the formula as

$$\boxed{\sum_{n=1}^{N}\sum_{k=1}^{K} r_{nk} \log p\left(\mathbf{x}_n \mid \boldsymbol{\theta}_k\right) + \sum_{n=1}^{N}\sum_{k=1}^{K} r_{nk} \log \varphi_k}$$

Solve the M-step – Mixture Models

$$\sum_{n=1}^{N}\sum_{k=1}^{K} r_{nk} \log p\left(\mathbf{x}_n \mid \boldsymbol{\theta}_k\right) + \sum_{n=1}^{N}\sum_{k=1}^{K} r_{nk} \log \varphi_k$$

The parameters to estimate: $\left\{\boldsymbol{\theta}_k, \varphi_k\right\}$ for all k = 1….K

- As r_{nk} has been computed in the E-step (fixed), the first term involves the pdf of the K components and the second their priors
- Update the parameters $\left\{\boldsymbol{\theta}_k\right\}$ of the components: maximize the first term
- Update the priors $\left\{\varphi_k\right\}$ for the K components: maximize the second term

Solve the M-step – Mixture Models

To derive an update for the k-th prior φ_k, we need to maximize

$$\sum_{n=1}^{N}\sum_{k=1}^{K} r_{nk}\log\varphi_k, \quad s.t., \sum_{k=1}^{K}\varphi_k = 1$$

We use a Lagrange multiplier λ to absorb the constraint into the objective function (see next page):

$$\sum_{n=1}^{N}\sum_{k=1}^{K} r_{nk}\log\varphi_k - \lambda(\sum_{k=1}^{K}\varphi_k - 1)$$

Taking the derivatives wrt φ_k and λ we get

$$\frac{1}{\varphi_k}\sum_{n=1}^{N} r_{nk} - \lambda = 0$$

$$\sum_{k=1}^{K}\varphi_k - 1 = 0$$

$$\lambda = \sum_{n=1}^{N}\sum_{k=1}^{K} r_{nk} = N$$

$$\varphi_k = \frac{1}{N}\sum_{n=1}^{N} r_{nk}$$

Generalized Lagrange

Refer to Chapter 5 of the book “Convex Optimization”

https://web.stanford.edu/~boyd/cvxbook/bv_cvxbook.pdf

We consider an optimization problem in the standard form (4.1):

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \ldots, m \\ & h_i(x) = 0, \quad i = 1, \ldots, p, \end{array} \tag{5.1}$$

with variable $x \in \mathbf{R}^n$. We assume its domain $\mathcal{D} = \bigcap_{i=0}^m \mathbf{dom}\, f_i \ \cap \ \bigcap_{i=1}^p \mathbf{dom}\, h_i$ is nonempty, and denote the optimal value of (5.1) by $p^\star$. We do not assume the problem (5.1) is convex.

The basic idea in Lagrangian duality is to take the constraints in (5.1) into account by augmenting the objective function with a weighted sum of the constraint functions. We define the *Lagrangian* $L : \mathbf{R}^n \times \mathbf{R}^m \times \mathbf{R}^p \to \mathbf{R}$ associated with the problem (5.1) as

$$L(x, \lambda, \nu) = f_0(x) + \sum_{i=1}^m \lambda_i f_i(x) + \sum_{i=1}^p \nu_i h_i(x),$$

with $\mathbf{dom}\, L = \mathcal{D} \times \mathbf{R}^m \times \mathbf{R}^p$. We refer to λ_i as the *Lagrange multiplier* associated with the ith inequality constraint $f_i(x) \leq 0$; similarly we refer to ν_i as the Lagrange multiplier associated with the ith equality constraint $h_i(x) = 0$. The vectors λ and ν are called the *dual variables* or *Lagrange multiplier vectors* associated with the problem (5.1).

Solve the M-step – Mixture Models

To derive an update for the parameters of the mixture distributions, we need to maximize the first term:

$$\sum_{n=1}^{N} \sum_{k=1}^{K} r_{nk} \log p\left(\mathbf{x}_n \mid \boldsymbol{\theta}_k\right)$$

We now need explicit distributions to expand our formulation.

Gaussian Mixture Models (GMM)

For GMM, we assume the mixture components follow the multivariate Gaussian distributions, i.e.,

$$p\left(\mathbf{x}_n \mid \boldsymbol{\theta}_k\right) = \frac{1}{(2\pi)^{d/2}\left|\boldsymbol{\Sigma}_k\right|^{1/2}} \exp\left(-\frac{1}{2}\left(\mathbf{x}_n - \boldsymbol{\mu}_k\right)^{\mathsf{T}} \boldsymbol{\Sigma}_k^{-1}\left(\mathbf{x}_n - \boldsymbol{\mu}_k\right)\right)$$

where the parameters $\boldsymbol{\theta}_k = \left(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k\right)$

Then the first term becomes

$$\sum_{n=1}^{N}\sum_{k=1}^{K} r_{nk}\left(-\frac{1}{2}\ln(2\pi)^d\left|\boldsymbol{\Sigma}_k\right| - \frac{1}{2}\left(\mathbf{x}_n - \boldsymbol{\mu}_k\right)^{\mathsf{T}} \boldsymbol{\Sigma}_k^{-1}\left(\mathbf{x}_n - \boldsymbol{\mu}_k\right)\right)$$

Gaussian Mixture Models (GMM)

Taking the partial derivative wrt μ_k and setting it to 0, we get

$$\sum_{n=1}^{N} r_{nk} \Sigma_k^{-1} \left(\mathbf{x}_i - \mu_k \right) = \mathbf{0}$$

from which we can easily solve for μ_k to obtain the updated estimate

$$\mu_k^{new} = \frac{\sum_{n=1}^{N} \mathbf{x}_n r_{nk}}{\sum_{n=1}^{N} r_{nk}}$$

Gaussian Mixture Models (GMM)

Taking the derivative with respect to Σ_k^{-1} and setting it to 0, we get

$$\sum_{n=1}^{N} r_{nk} \left(\frac{1}{2}\Sigma_k - \frac{1}{2}\left(\mathbf{x}_n - \mu_k^{new}\right)\left(\mathbf{x}_n - \mu_k^{new}\right)^{\mathsf{T}} \right) = 0$$

using basic Linear Algebra: $\frac{\partial\left(x^{\mathsf{T}} A y\right)}{\partial A} = x y^{\mathsf{T}}$ $\quad \frac{\partial(\ln|A|)}{\partial A} = \left(A^{\mathsf{T}}\right)^{-1}$ $\quad \left|A^{-1}\right| = 1/|A|$

(see The Matrix Cookbook)

from which we can solve for Σ_k to obtain the updated estimate

$$\Sigma_k^{new} = \frac{\sum_{n=1}^{N} r_{nk}\left(\mathbf{x}_n - \mu_k^{new}\right)\left(\mathbf{x}_n - \mu_k^{new}\right)^{\mathsf{T}}}{\sum_{n=1}^{N} r_{nk}}$$

Gaussian Mixture Models (GMM)

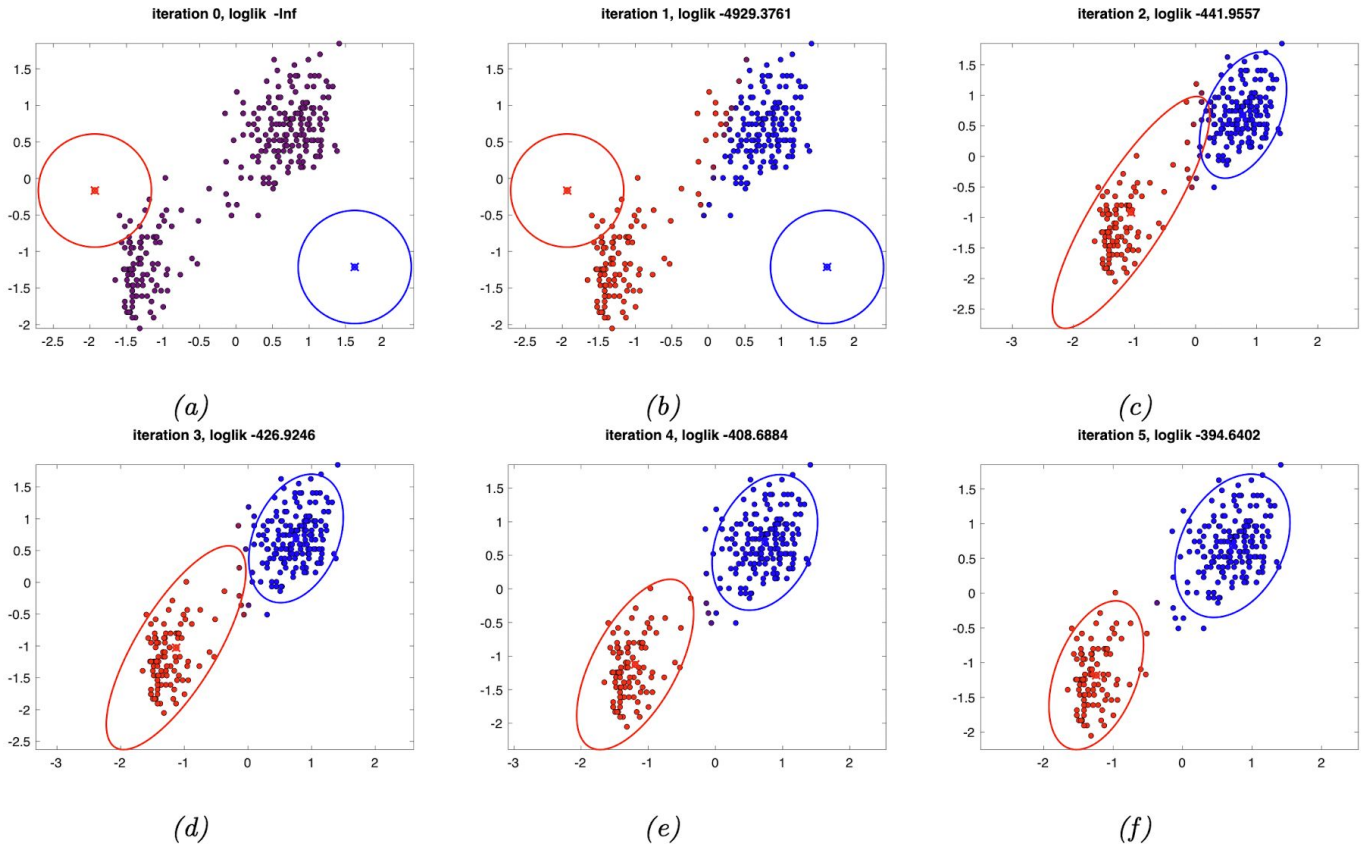
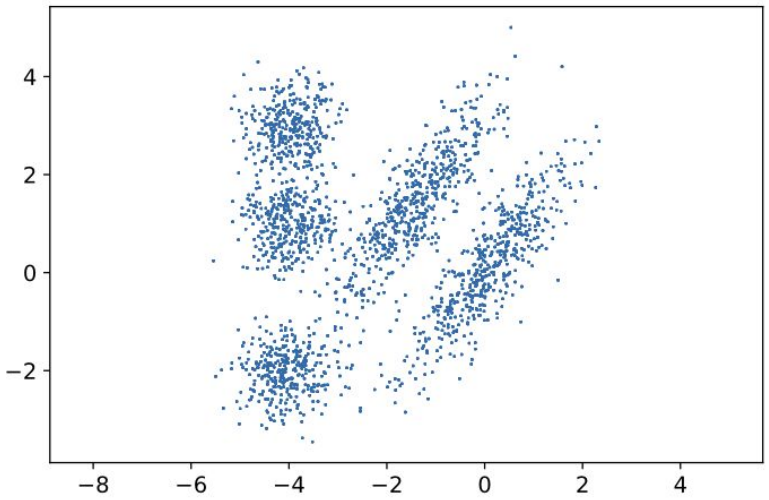


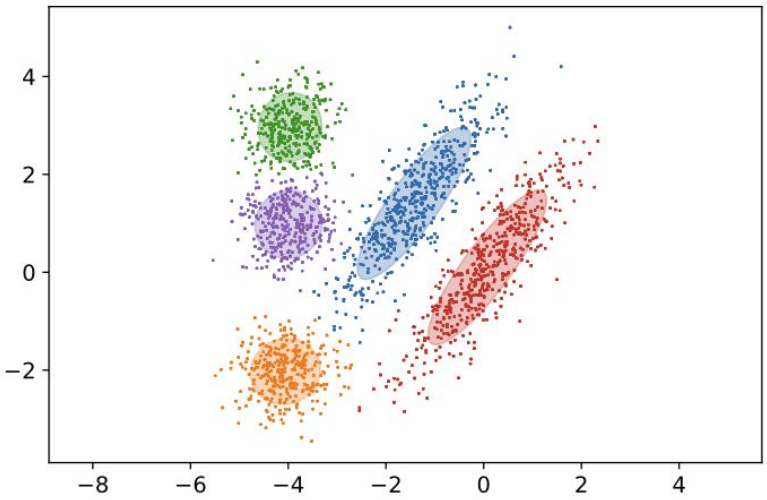
Illustration of the EM for a GMM for the first 6 steps. The degree of redness indicates the degree to which the point belongs to the red cluster, and similarly for blue; thus purple points have a roughly 50/50 split in their responsibilities to the two clusters.

From C. Bishop. Pattern recognition and machine learning. 2006

Gaussian Mixture Models (GMM)



(a)



(b)

a) Some data in 2d. (b) A possible clustering using K = 5 clusters computed using a GMM

Relationship between hard clustering and soft clustering

Proposition: K-means (hard clustering) is a special case of Mixture Models (soft clustering)

Proof: The quantity r_{nk} is called the responsibility of cluster k for the n-th data point. Given the responsibilities, we can compute the most probable cluster assignment as follows:

$$\hat{z}_n = \arg\max_k r_{nk} = \arg\max_k \left[\log p\left(\boldsymbol{x}_n \mid z_n = k, \boldsymbol{\theta}\right) + \log p\left(z_n = k \mid \boldsymbol{\theta}\right)\right]$$

This is hard clustering (use hard EM). If we use the responsibilities to fractionally assign each data point to different clusters, it is called soft clustering (use soft EM).

If we have a uniform prior over z_n, and we use spherical Gaussians with $\Sigma_k = \mathbf{I}$, the hard clustering problem reduces to

$$\hat{z}_n = \operatorname*{argmin}_k \|\boldsymbol{x}_n - \hat{\boldsymbol{\mu}}_k\|_2^2$$

In this way, we assign each data point to its closest centroid, as measured by Euclidean distance. This is the basis of the K-means clustering algorithm.

Part 7: EM for Document Clustering

EM for Document Clustering

Document Clustering is another application of Mixture Models. It is very similar to GMMs. This scenario involves two levels of random variables, namely documents and words. We can also use EM to solve MLE with hidden variables.

GMM: use a **continuous** Gaussian density to model a cluster

Document Clustering: use a **discrete** probability distribution (probability mass function), specifically the multinomial distribution, to represent a cluster

Document Clustering

- Given a collection of documents $\{d_1, d_2, \ldots, d_N\}$, we would like to partition them into K clusters.
- Document representation
  - Each document is made of some text
  - We treat a document as a *set* of words in its text irrespective of their positions
  - Also, we assume the words appearing in our collection of documents come from a shared dictionary denoted by $\mathcal{A}$

Understanding the denotations: an example

- d_1=this one has a little star
- d_2=this one has a little car
- d_3=I would not like them
- d_4=I would not like them anywhere
- d_5=I would like eggs

A = {this, one, has, a, little, star, car, I, would, not, like, them, there, anywhere, eggs}

- Assume we know the clusters beforehand (in reality we don't)
  - K=2 (two clusters from two books)
  - C_1=(d_1, d_2), C_2=(d_3, d_4, d_5)
  - $\varphi_1 = 0.4$ (2/5), $\varphi_2 = 0.6$ (3/5)
  - Words for C_1=(this, one, has, a, little, star, car)
  - μ_1=(2/12, 2/12, 2/12, 2/12, 1/12, 1/12, 0, 0, 0, 0, 0, 0, 0, 0)
  - Words for C_2=(I, would, not, like, them, anywhere, eggs)
  - μ_2=(0, 0, 0, 0, 0, 0, 0, 0.2, 0.2, 2/15, 0.2, 2/15, 1/15, 1/15)

Document Clustering

- For each document d_{n}
  - Toss the K-face dice (with the probability parameter φ) to choose the face k (i.e., the cluster) that the nth document belongs to
  - For each word w in the document d_{n}
    - > Generate the word by tossing the dice (with the probability parameter μ_{kw}) corresponding to the face k

Parameters:

- The proportion of each cluster (mixing coefficients)
- The proportion of each word (given a cluster k, how likely we can generate this word)
- The constraints (sum of probabilities is 1) allow us to use Lagrange multiplier to learn the parameters

Likelihood

$\mathcal{A}$ ={this, one, has, little, star, car, I, would, not, like, them, here, there, anywhere, do, green, eggs, ham}

shared dictionary for all documents

P(this, one, has, little, little, star) = P(this)P(one)P(has) P(little) P(little)P(star)

words in the document are independent; non-autoregressive

$$P(d \mid k) = \prod_{w \in d} P(w \mid k) = \prod_{w \in \mathcal{A}} P(w \mid k)^{c(w,d)}$$

where c(w, d) is the number of occurrences of the word w in the document d.

Likelihood of Complete Data

We assume a document and its cluster label are known (observable). Then we can directly write out and solve MLE regarding the joint likelihood of generating a document d and its cluster label k:

$$p(k, d) = p(k)p(d \mid k) = \varphi_k \prod_{w \in d} \mu_{kw} = \varphi_k \prod_{w \in \mathcal{A}} \mu_{kw}^{c(w,d)}$$

But in practice, the document cluster labels are not given to us, they are hidden variables.

Complete Data

Now assume we have many documents $\{d_1, d_2, \ldots, d_N\}$

We use variables $\mathbf{z}_n$ to denote the cluster assignments for the nth document

$$\mathbf{z}_n = (z_{n1}, z_{n2}, \ldots, z_{nK})$$

$$z_{nk} = \begin{cases} 1, & d_n \in \mathcal{C}_k \\ 0, & d_n \notin \mathcal{C}_k \end{cases}$$

Only one z_{nk} is 1. The rests are zero.

Complete Data

Likelihood of the complete data

$$p(d_1, \mathbf{z}_1, \ldots, d_N, \mathbf{z}_N) = \prod_{n=1}^{N} \prod_{k=1}^{K} \left(\varphi_k \prod_{w \in \mathcal{A}} \mu_{kw}^{c(w,d_n)} \right)^{z_{nk}}$$

$$z_{nk} = \{ \begin{array}{l} 1, d_n \in \mathcal{C}_k \\ 0, d_n \notin \mathcal{C}_k \end{array}$$

With the constraints that $\sum_{w \in \mathcal{A}} \mu_{kw} = 1$ and $\sum_{k=1}^{K} \varphi_k = 1$

We can use generalized Lagrange to optimize the parameters

Complete Data

Through introducing the Lagrange multiplier on the maximum likelihood function, we can solve the parameters by setting their gradients to 0:

Mixing coefficients: $\varphi_k = \frac{N_k}{N}$ where $N_k = \sum_{n=1}^{N} z_{nk}$

Word proportion parameters: $$\mu_{kw} = \frac{\sum_{n=1}^{N} z_{nk} c\left(w, d_n\right)}{\sum_{w' \in \mathcal{A}} \sum_{n=1}^{N} z_{nk} c\left(w', d_n\right)}$$

Incomplete Data and EM

We don't know the cluster labels of the documents and need to represent the incomplete-data likelihood as the complete-data likelihood marginalized over the hidden variables $\mathbf{Z}_n$. Note that this is also the formulation of the mixture model.

$$p\left(d_1, \ldots, d_N\right) = \prod_{n=1}^{N} p\left(d_n\right) = \prod_{n=1}^{N} \sum_{k=1}^{K} p\left(d_n, z_n = k\right) = \prod_{n=1}^{N} \sum_{k=1}^{K} \left(\varphi_k \prod_{w \in \mathcal{A}} \mu_{kw}^{c(w, d_n)} \right)$$

Likelihood of the incomplete data

- Hard to derive the analytical solutions
- Resort to EM algorithm for MLE parameter estimation with hidden variables

Incomplete Data and EM

E-step: estimate the distributions of hidden variables to get the tight lower bound function (Q function)

$$r\left(z_{nk}\right) = p\left(z_n = k \mid d_n, \boldsymbol{\theta}^{\text{old}}\right)$$

The notation $p(z_n = k)$ is equivalent to $p(z_{nk} = 1)$

Incomplete Data and EM

The M-step maximizes the following expected complete data log likelihood:

$$
\begin{aligned}
Q\left(\boldsymbol{\theta}, \boldsymbol{\theta}^{\text{old}}\right) &:= \sum_{n=1}^{N} \sum_{k=1}^{K} p\left(z_{nk}=1 \mid d_n, \boldsymbol{\theta}^{\text{old}}\right) \log p\left(z_{nk}=1, d_n \mid \boldsymbol{\theta}\right) \\
&= \sum_{n=1}^{N} \sum_{k=1}^{K} r\left(z_{nk}\right) \log \left(\varphi_k \prod_{w \in \mathcal{A}} \mu_{kw}^{c(w, d_n)}\right) \\
&= \sum_{n=1}^{N} \sum_{k=1}^{K} r\left(z_{nk}\right) \left(\log \varphi_k + \log \prod_{w \in \mathcal{A}} \mu_{kw}^{c(w, d_n)}\right) \\
&= \sum_{n=1}^{N} \sum_{k=1}^{K} r\left(z_{nk}\right) \left(\log \varphi_k + \sum_{w \in \mathcal{A}} c\left(w, d_n\right) \log \mu_{kw}\right)
\end{aligned}
$$

Incomplete Data and EM

$$\varphi_k = \frac{N_k}{N} \quad \text{where} \quad N_k = \sum_{n=1}^{N} r(z_{nk})$$

$$\mu_{kw} = \frac{\sum_{n=1}^{N} r\left(z_{nk}\right) c\left(w, d_n\right)}{\sum_{w' \in \mathcal{A}} \sum_{n=1}^{N} r\left(z_{nk}\right) c\left(w', d_n\right)}$$

EM for Document Clustering Summary

- Choose an initialization for the parameters $\boldsymbol{\theta}^0 = \left(\boldsymbol{\varphi}^0, \boldsymbol{\mu}_1^0, \ldots, \boldsymbol{\mu}_K^0\right)$
- While the convergence is not met, t ← t + 1:
  - E-step: Obtain $\forall n, \forall k : r\left(z_{nk}\right)$ based on $\boldsymbol{\theta}^t$
  - M-step: Obtain $\boldsymbol{\theta}^{\mathrm{t}+1}$ by solving MLE for the expected complete-data log likelihood

Other Methods for Document Clustering

- Use distance metrics (e.g., Euclidean distance, Cosine distance, Wasserstein distance in optimal transport, Manhattan distance, Hamming distance, etc) to measure the similarity between documents/cluster vectors
- Can use K-Means to cluster the documents into K clusters
- Spectral clustering

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Thank you